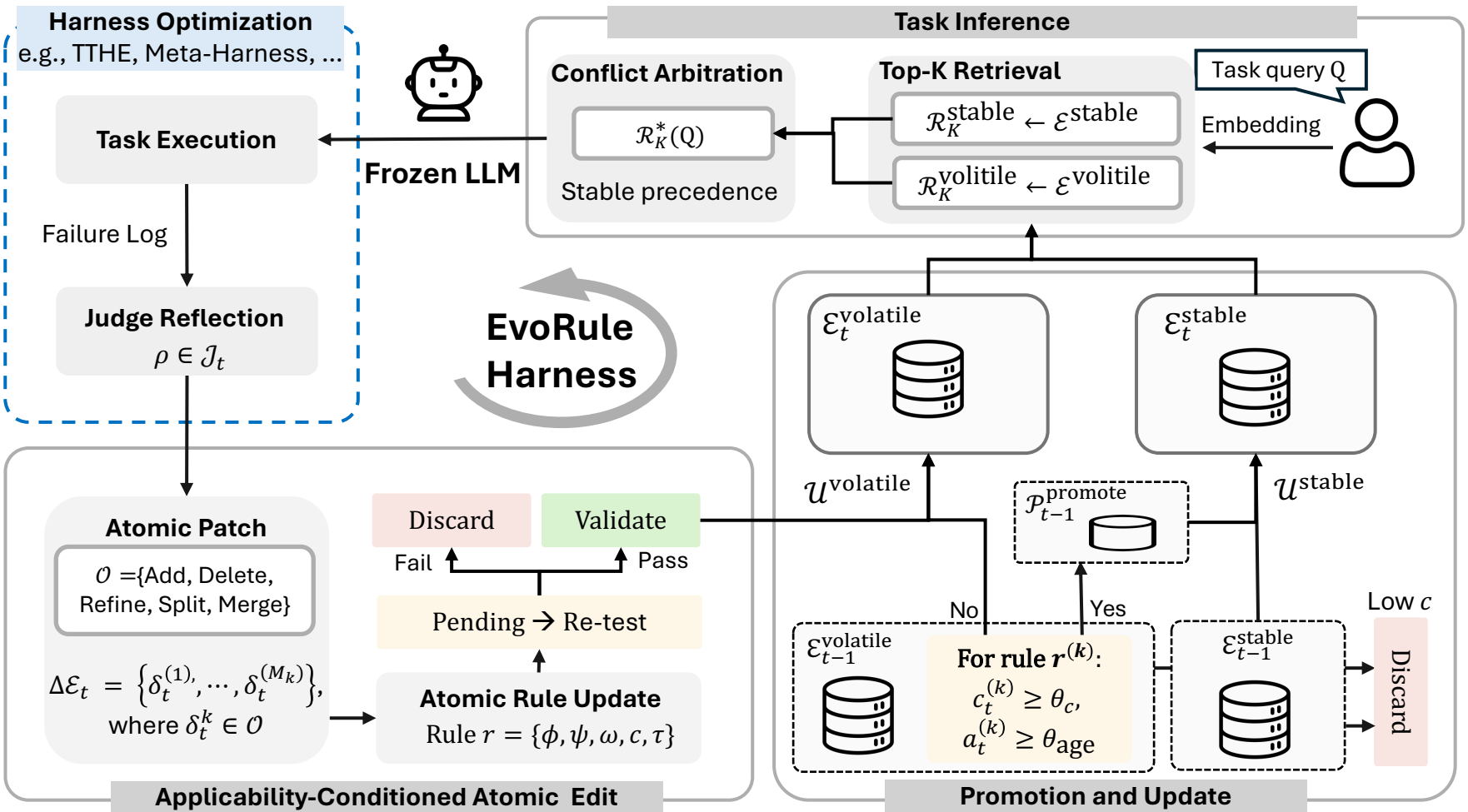

Harness Optimization
e.g., TTHE, Meta-Harness, ...
Task Execution
Failure Log
Judge Reflection
$\rho \in \mathcal{J}_t$
Frozen LLM
Task Inference
Conflict Arbitration
$\mathcal{R}_K^*(Q)$
Stable precedence
Top-K Retrieval
$\mathcal{R}_K^{\text{stable}} \leftarrow \mathcal{E}^{\text{stable}}$
$\mathcal{R}_K^{\text{volitile}} \leftarrow \mathcal{E}^{\text{volitile}}$
Task query Q
Embedding
EvoRule Harness
$\mathcal{E}_t^{\text{volatile}}$
$\mathcal{E}_t^{\text{stable}}$
$\mathcal{U}^{\text{volatile}}$
$\mathcal{P}_{t-1}^{\text{promote}}$
$\mathcal{U}^{\text{stable}}$
Atomic Patch
$\mathcal{O}$ ={Add, Delete, Refine, Split, Merge}
$\Delta\mathcal{E}_t = \{\delta_t^{(1)}, \cdots, \delta_t^{(M_k)}\}$, where $\delta_t^k \in \mathcal{O}$
Discard
Validate
Fail
Pass
Pending → Re-test
Atomic Rule Update
Rule $r = \{\phi, \psi, \omega, c, \tau\}$
No
Yes
Low c
$\mathcal{E}_{t-1}^{\text{volatile}}$
For rule $r^{(k)}$:
$c_t^{(k)} \geq \theta_c$,
$a_t^{(k)} \geq \theta_{\text{age}}$
$\mathcal{E}_{t-1}^{\text{stable}}$
Discard
Applicability-Conditioned Atomic Edit
Promotion and Update